

KUSONG, Korea, Democratic People's Republic of

WMO: 470370

Lat: 39.983N

Lon: 125.250E

Elev: 99

StdP: 100.14

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.6	-14.7	-23.7	0.4	-13.2	-21.9	0.5	-12.0	7.2	-9.7	6.3	-9.4	1.1	320	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	31.6	24.9	30.3	23.9	29.1	23.4	26.5	29.9	25.5	28.5	24.7	27.3	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.1	28.6	24.6	19.9	27.2	23.9	19.0	26.4	83.8	30.0	79.2	28.6	75.9	27.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.0	4.2	-19.5	34.1	2.4	1.2	-21.2	35.0	-22.7	35.7	-24.0	36.3	-25.7	37.2		
(5)			-19.8	27.8	2.4	1.4	-21.5	28.8	-22.9	29.6	-24.3	30.4	-26.0	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-6.5	-3.5	2.8	10.1	16.6	21.4	24.4	24.6	19.6	12.5	3.2	-4.8		
	DBStd	11.57	3.94	4.01	3.67	3.69	2.94	2.11	2.03	2.21	2.69	3.65	4.82	4.30		
	HDD10.0	1840	510	377	223	43	1	0	0	0	0	18	209	459		
	HDD18.3	3550	768	611	481	248	69	2	0	1	17	182	455	717		
	CDD10.0	1882	0	0	1	46	206	342	446	453	288	96	4	0		
	CDD18.3	549	0	0	0	0	16	94	188	195	56	2	0	0		
	CDH23.3	3821	0	0	0	8	181	613	1239	1431	333	15	0	0		
	CDH26.7	1018	0	0	0	0	31	146	352	441	47	1	0	0		
(14) Wind	WSAvg	1.3	1.4	1.5	1.4	1.5	1.3	1.1	1.1	1.0	1.0	1.1	1.3	1.4		
(15) Precipitation	PrecAvg	1216	8	21	25	57	93	125	377	286	111	54	48	17		
	PrecMax	2069	27	95	81	115	210	224	1057	704	305	124	135	57		
	PrecMin	848	0	1	0	2	11	42	125	69	19	4	7	2		
	PrecStd	312	8	23	16	31	49	55	182	163	80	32	38	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	9.0	16.7	24.6	29.6	31.6	34.0	33.7	29.8	25.2	17.1	6.8		
		MCWB	2.1	3.8	10.4	15.1	19.2	22.0	26.7	26.7	22.6	17.9	12.0	3.9		
	2%	DB	2.6	6.4	13.6	21.8	27.2	29.9	31.6	31.8	28.0	22.5	13.8	4.6		
		MCWB	-0.1	2.6	8.3	13.8	17.9	21.4	25.4	25.9	21.5	16.4	10.4	2.5		
	5%	DB	1.2	4.4	11.4	19.6	25.2	28.3	30.0	30.5	26.7	20.6	12.0	2.9		
		MCWB	-1.0	1.1	6.7	12.8	17.1	20.9	24.6	24.6	20.4	15.2	8.6	0.9		
	10%	DB	-0.3	2.6	9.2	17.4	23.4	26.9	28.6	29.2	25.3	18.8	10.2	1.4		
		MCWB	-2.5	-0.2	5.1	11.1	16.2	20.4	24.2	24.0	19.7	14.0	7.7	-0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	5.4	11.0	16.6	20.7	24.1	28.2	28.4	24.2	19.1	12.9	5.1		
		MCDB	4.4	7.9	15.2	21.9	27.2	29.3	32.5	32.2	27.8	23.4	15.8	5.9		
	2%	WB	0.5	3.1	9.1	14.8	19.2	23.0	26.6	26.8	22.7	17.3	11.0	2.7		
		MCDB	2.0	6.0	12.8	20.2	24.8	27.4	29.9	30.3	26.1	21.2	13.2	4.2		
	5%	WB	-0.9	1.4	7.2	13.4	18.2	22.1	25.6	25.7	21.6	16.1	9.5	1.2		
		MCDB	1.0	3.7	10.9	18.5	23.5	26.4	28.5	28.9	25.3	19.6	11.4	2.6		
	10%	WB	-2.4	0.2	5.4	11.9	17.1	21.3	24.8	25.0	20.7	14.8	7.9	-0.2		
		MCDB	-0.5	2.2	8.8	16.5	22.0	25.3	27.3	27.6	24.3	17.9	9.8	1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.5	10.0	11.6	11.7	9.5	6.8	8.1	10.6	11.3	8.7	8.3		
		MCDBR	10.5	11.0	13.2	14.9	14.9	13.0	10.2	10.2	12.2	13.1	10.9	8.7		
	5% WB	MCWBR	8.3	7.9	8.9	8.5	7.5	6.4	5.3	5.2	6.5	7.7	8.0	7.0		
		MCWBR	9.7	10.0	12.4	13.6	12.9	10.3	8.0	7.9	10.0	11.6	9.7	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.394	0.471	0.534	0.615	0.641	0.716	0.689	0.584	0.511	0.537	0.478	0.407		
	taud		1.973	1.827	1.739	1.629	1.620	1.547	1.647	1.864	1.970	1.804	1.875	1.970		
	Ebn at Noon		715	707	711	686	681	632	644	704	728	635	613	660		
	Edh at Noon		128	172	209	249	256	276	248	194	164	174	142	120		
(44) All-Sky Solar Radiation	RadAvg		2.45	3.21	4.36	5.00	5.63	5.13	4.23	4.50	4.43	3.38	2.30	2.06		
	RadStd		0.14	0.22	0.31	0.39	0.47	0.57	0.51	0.53	0.36	0.22	0.24	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page